

Modeling Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

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Thesis Presentation

Modeling Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

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1. Supervisor - Martin Lindström
2. Examiner - Ragnar Thobaben
3. Opponent - Divyayan Day

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- 1 Patients suffering from critical illnesses incur metabolic changes from acute stress

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└ Stress Hyperglycaemia

Stress Hyperglycaemia

Patients suffering from critical illnesses incur metabolic changes from acute stress

1. Inflammatory response secretes stress hormones
2. Acute myocardial infarction, sepsis, stroke

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- ❏ Instances of stress-induced hyperglycaemia have been identified to accompany the onset of *mortality*
- ❏ In a controlled study, tightly maintaining glycaemic levels (4 mmol/L - 6 mmol/L) reduced mortalities by 50% compared to conventional methods

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Limitations of Intensive Insulin Therapy

- 1 Relying on insulin infusions to tightly control blood glucose struggles to address patient variability

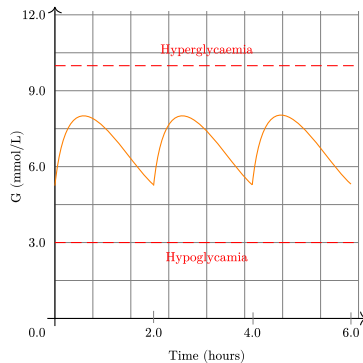


Figure: Blood Glucose Example



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Limitations of Intensive Insulin Therapy

1. Infrequent measurements, not considering nutrition, and not complex to monitor dynamics changes in patient health.

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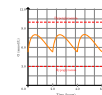


Figure: Blood Glucose Example

Limitations of Intensive Insulin Therapy

- 1 Relying on insulin infusions to tightly control blood glucose struggles to address patient variability
- 2 Consequences in poor patient adaptability has resulted in greater cases of hypoglycaemia and high glycaemic variation

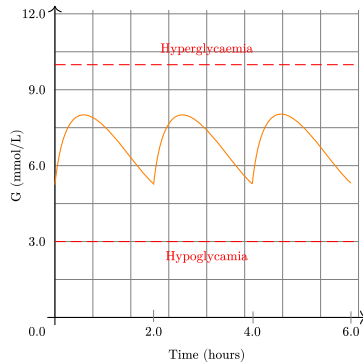


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Limitations of Intensive Insulin Therapy

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2. A patient with nutrition intake at 2 and 4 hours

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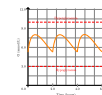


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